

Ryan Curry

Theoretical Division (T-2)
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Employment

- 07/2026 – **Postdoctoral Research Associate**
Theoretical Division (T-2), Los Alamos National Laboratory, Los Alamos, NM
- 09/2021 – **Graduate Teaching & Research Assistant**
06/2026 *Department of Physics, University of Guelph, Guelph, ON*
- 06/2024 – **Graduate Research Assistant - Internship**
09/2024 *Theoretical Division (T-2), Los Alamos National Laboratory, Los Alamos, NM*
- 06/2019 – **Undergraduate Research Assistant**
09/2021 *Department of Physics, University of Guelph, Guelph, ON*

Education

- 09/2023– **Ph.D in Physics**, University of Guelph
06/2026 *Microscopic Nuclear Physics with Quantum Monte Carlo: Perturbation, Discretization, and Emulation*
- 09/2021– **M.Sc. in Physics**, University of Guelph
08/2023 *Quantum Monte Carlo for Nuclear Physics: On and Off the Lattice*
- 09/2017– **B.Sc (Honours) in Physics**, University of Guelph
05/2021

Honors & Awards

- 2025-2027 **NSERC Postgraduate Scholarship - Doctoral**
Natural Sciences and Engineering Research Council of Canada
(Declined from Fall 2026 onwards to leave for postdoctoral position)
- 2025 **Teaching Assistant Excellence Award**
University of Guelph, College of Engineering and Physical Sciences
- 2024 **Governor General's Gold Medal**
University of Guelph, on behalf of the Governor General
- 2024 **Ontario Graduate Scholarship - Masters**
Ministry of Training, Colleges and University & University of Guelph
- 2023 **Highdale Farms Graduate Scholarship in Natural Sciences**
University of Guelph through donors Arthur and Rosmarie Spoerri
- 2021-2027 **Graduate Tuition Scholarship - M.Sc (2021) and Ph.D. (2023-2027)**
Department of Physics at the University of Guelph (declined from 2024 onward due to holding OGS and PGS-D awards)
- 2021 **J.B. Reynolds Graduation Medal**
Department of Physics at the University of Guelph
- 2021 **MacNaughton Scholarship**
Department of Physics at the University of Guelph

Grants & Allocations

- 2025 - 2026 **Co-Investigator**
Exploring local chiral effective field theory interactions with Quantum Monte Carlo methods
National Energy Research Scientific Computing Center (NERSC) Grant
6000 Node-hours in current granting period

Publications

Underlining names of students under my direct or co-supervision.

- [9] **Exploring the shell structure of trapped superfluid gases**
Georgios Palkanoglou, Ryan Curry, and Stefano Gandolfi
Submitted to Physical Review C
[arXiv:2607.14214](#)
- [8] **Conformal prediction for uncertainties in the neutron star equation of state**
Habib Yousefi Dezdaran, Ryan Curry, Cassandra L. Armstrong and Alexandros Gezerlis
Submitted to Physical Review C
[arXiv:2604.21039](#)
- [7] **Quantum Monte Carlo Calculations of Light Nuclei with Fully Propagated Theoretical Uncertainties**
Ryan Curry, Kai Hebeler, Stefano Gandolfi, Alexandros Gezerlis, Achim Schwenk, Rahul Somasundaram, and Ingo Tews
[Phys. Lett. B, **881**, 140908 \(2026\)](#)
[arXiv:2510.15860](#)
- [6] **Ab initio study of the neutron and Fermi polarons on the lattice**
Ryan Curry, Jasmine Kozar, and Alexandros Gezerlis
[Phys. Rev. C, **113**, 044002 \(2026\)](#)
[arXiv:2510.05233](#)
- [5] **Conformal prediction for uncertainties in nucleon-nucleon scattering**
Habib Yousefi Dezdaran, Ryan Curry, and Alexandros Gezerlis
[Physical Review C, **113**, 014004](#)
[arXiv:2507.08085](#)
- [4] **Perturbative treatment of non-local chiral interactions in auxiliary-field diffusion Monte Carlo calculations**
Ryan Curry, Rahul Somasundaram, Stefano Gandolfi, Alexandros Gezerlis, and Ingo Tews
[Physical Review C, **111**, 015801 \(2025\)](#)
[arXiv:2409.16365](#)
- [3] **BCS-BEC Crossover of the strongly interacting ^6Li - ^{40}K mixture**
Stefano Gandolfi, Ryan Curry, and Alexandros Gezerlis
[Physical Review A, **110**, 043320 \(2024\)](#)
[arXiv:2407.12102](#)
- [2] **Auxiliary Field Quantum Monte Carlo for Dilute Neutrons on the Lattice**
Ryan Curry, Jayani Dissanayake, Stefano Gandolfi, and Alexandros Gezerlis
[Philosophical Transactions of the Royal Society A, **382**, 20230127 \(2024\)](#)
[arXiv:2310.01504](#)
- [1] **Second-Order Perturbation Theory in Continuum Quantum Monte Carlo Calculations**
Ryan Curry, Joel E. Lynn, Kevin E. Schmidt, and Alexandros Gezerlis
[Physical Review Research, **5**, L042021 \(2023\)](#)
[arXiv:2302.07285](#)

Invited Talks

- 2027-05 (Upcoming) *Emulators for constraining chiral EFT Hamiltonians*
Workshop on Bridging Effective Field Theory Across Subatomic Scales
ECT*, Trento, Italy

- 2026-09 (Upcoming) *Uncertainty quantification in many-body physics: from light nuclei to neutron stars*
Physics Department Seminar
University of New Mexico, Albuquerque, NM, USA
- 2026-05 *QMC and Chiral EFT: Uncertainty Quantification and Perturbation Theory*
Nuclear Hamiltonians for Advancing Nuclear Physics and Beyond Workshop
Institute for Nuclear Theory, Seattle, WA, USA
- 2026-02 *Emulators for uncertainty propagation: from nuclear forces to many-body physics*
2026 Workshop on Progress in Ab Initio Nuclear Theory
TRIUMF, Vancouver, BC, Canada
- 2025-02 *Perturbative treatment of non-local operators in quantum Monte Carlo*
2025 Workshop on Progress in Ab Initio Nuclear Theory
TRIUMF, Vancouver, BC, Canada
- 2024-08 *Perturbative treatment of non-local operators in an ab initio context*
T-2 Seminar
Theoretical Division, Los Alamos National Laboratory, Los Alamos, NM, USA
- 2024-05 *Perturbing the Envelope: A new approach to perturbation theory in quantum Monte Carlo*
7th Annual Meeting of the Toronto Ultra Cold Atoms Network (TUCAN)
McMaster University, Hamilton, ON, Canada
- 2023-08 *Novel pairing and testing EFT interactions in neutron matter and nuclei*
Nuclear Theory Seminar
Facility for Rare Isotope Beams (FRIB), Michigan State University, East Lansing, Michigan

Contributed Talks and Posters

- 2024-10 *Perturbative treatment of non-local operators in an ab initio context*
Contributed Talk, 2024 American Physical Society (APS) Division of Nuclear Physics (DNP) Meeting
Boston MA
- 2024-07 *Perturbing the Envelope: A new approach to perturbation theory in quantum Monte Carlo*
2024 T-Division Student Lightning Talks
Theoretical Division, Los Alamos National Laboratory, Los Alamos NM
- 2024-05 *Probing the nuclear many-body problem with quantum Monte Carlo*
Contributed Talk, 2024 Canadian Association of Physicists (CAP) Congress
Western University, London ON
- 2024-05 *Second order perturbation theory in quantum Monte Carlo*
Contributed Talk, Theory Canada 16 Conference
Institute for Quantum Computing (IQC), Waterloo, ON
- 2023-10 *Second-Order Perturbation Theory in Nuclear Quantum Monte Carlo*
Poster, 2023 College of Engineering and Physical Science Research Day
University of Guelph, Guelph, ON
- 2023-06 *Second-Order Perturbation Theory in Continuum Quantum Monte Carlo*
Contributed Talk, 2023 Canadian Association of Physicists (CAP) Congress
University of New Brunswick, Fredericton, NB
- 2023-05 *A Quantum Monte Carlo Study: From the Hubbard Model to The Unitary Fermi Gas*
Poster, 6th Annual Meeting of the Toronto Ultra Cold Atoms Network (TUCAN)
University of Toronto, Toronto, ON

2023-02 *How to consistently use modern nuclear forces in an ab initio technique*,
Contributed Talk, 60th Winter Nuclear and Particle Physics Conference
Banff, AB

Teaching

2022-2026 **Teaching Assistant, Department of Physics, University of Guelph**
— **Graduate Courses**
PHYS*7030: Quantum Field Theory
1 Semester: Fall 2025
PHYS*7730: Graduate Special Topics - Computational Methods in Physics
2 Semesters: Fall 2022, Fall 2023

— **Undergraduate Courses**
PHYS*3510/4500: Intermediate/Advanced Physics Laboratory
4 Semesters: Winter 2023, Fall 2023, Winter 2024, Winter 2025
PHYS*1130: Physics with Applications
1 Semester: Fall 2022
PHYS*1010: Introductory Electricity & Magnetism
1 Semester: Winter 2022

Students Supervised / Co-Supervised

2026 - Jim A. Papadopoulos (U. Guelph), Master's Student
Project: *Perturbative approaches to fitting chiral EFT interactions*

2025 - Jasmine Kozar (U. Guelph), Undergraduate Research Assistant
Project: *Building a renormalizable local chiral EFT interactions at leading order*
Project: *Emulators for lattice auxiliary-field quantum Monte Carlo*
Published in [Phys. Rev. C](#), **113**, 044002 (2026)

2025 - Isaiah Bennett (U. Guelph), Master's Student
Project: *Emulators for phenomenological calculations of deformed heavy nuclei*

2025 - Safwan Khan (University of Guelph), Master's Student
Project: *Lattice QMC with a one-body spin-orbit interaction*

2024–2026 Obada Al-Ajeh (U. Guelph), Master's Student
Project: *Weinberg eigenvalue analysis of local chiral effective field theory interactions*

2024 Olivia Parker (U. of Guelph), Fourth Year Undergraduate Research Project
Project: *Study of many-body systems of polarized, strongly interacting fermions*

2024 Matthew Burrows-Balka (U. of Guelph), Fourth Year Undergraduate Research Project
Project: *Shape independence of microscopic potentials in the nuclear bound state*

Referee Activity

Since 2026 Physics Letters B, European Physical Journal A
Since 2025 Springer: Scientific Reports, Discover Applied Sciences

Professional Service

2026 **Judge for Student Lightning Talk Competition**, Theoretical Division, Los Alamos National Laboratory
- Judged 10 minute talk from graduate, undergraduate, and high school students working in T-division as Summer interns.

- 2025 **Served on department search committee for Assistant Faculty position**, University of Guelph/TRIUMF
- Reviewed applicant packages and help craft long/short list of candidates
 - Interviewed potential candidates / Voted on final candidate.
- 2023-2026 **Graduate Representative for Physics Department Undergrad Curriculum Committee**
- Attended quarterly meetings to discuss/implement strategies for improving the undergraduate physics curriculum at the University of Guelph.
 - Conducted focus groups with graduate students to learn more about their undergraduate experience, delivered report on findings.
- 2022-2026 **Co-founded and ran the Guelph Graduate Physics Society**, University of Guelph
- Reviving and running a weekly graduate seminar series that allowed graduate students to present their research.
 - Securing funding and organization for the revitalization of Physics Graduate Student Lounge
 - Assisted with the organization and running of ongoing events for physics graduate students.
- 2023 **Organized and ran the College of Engineering and Physical Sciences Undergraduate Poster Session**
- Responsible for all administration of Undergraduate STEM poster session.
 - Collected and organized poster contributions from several departments (Physics, Chemistry, Biology, Engineering, Computer Science, Math).
 - Organized and supervised graduate student judges.
- 2022 **Judge for 58th Annual Canadian Undergraduate Physics Conference (CUPC)**
- Responsible for judging undergraduate research presentation in the nuclear physics sessions.
- 2021-present **Volunteer for Department Outreach Events**
- Volunteered for various Department of Physics outreach events that are aimed at attracting new undergraduate students to the department, University of Guelph